<u>**EE 522 - PsiQuantum Group - Weekly Report 4**</u>

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Previous Meeting Items

We began implementing algorithms in code. One algorithm had previously been implemented and tested in a different quantum software suite.

Future Plan

The one algorithm (TTK) is implemented in the PsiQuantum Workbench platform. The other two algorithms are in advanced stages of implementations in Workbench. We will implement a testing apparatus that's common to all of us. The exact signatures of our respective algorithms will be different, for example whether or not for *N* qubits we need an additional register of *N* qubits to store the solution. But there should be enough commonalities to motivate a common suite of underlying tests. It was emphasized in particular that "uncomputing" auxiliary qubits is important as it can create subtle, obnoxious problems.

Problems

It was emphasized that these algorithm implementations should be written with an end user in mind, such as with regards to complete documentation (e.g. are the input registers overwritten with the result or is there a separate register returned) and naming standards like [Algorithm Name]Adder.py for our qubricks.